Paolo Giaretta

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Education

- **Massachusetts Institute of Technology (MIT)** — Cambridge, MA, USA
  Master's Thesis Student – Laboratory for Information and Decision Systems (LIDS) — Feb. 2026 – Present
  — Working with Prof. Navid Azizan on safe and reliable machine learning for robotic applications, with a focus on hard-constrained diffusion sampling.
  — Publication: *DiRecT: Safe Diffusion-Based Planning via Receding-Horizon Denoising, P. Giaretta, Z. Li, N. Azizan* [Under review at NeurIPS 2026].
- **École Polytechnique Fédérale de Lausanne (EPFL)** — Lausanne, Switzerland
  M.Sc. Mechanical Engineering and Data Science – 5.95/6.0 GPA (top 1%) — Sep. 2023 – Sep. 2026
  — Coursework: Deep Learning, Reinforcement Learning, Convex Optimization, Applied Machine Learning.
  — Teaching Assistant for the courses *Machine Learning Programming* and *Numerical Flow Simulation*.
- **Politecnico di Milano** — Milan, Italy
  B.Sc. Aerospace Engineering – 110 cum laude/110, 29.74/30 GPA (top 1%) — Sep. 2023 – Sep. 2026

Experience

- **European Space Agency** — Noordwijk, Netherlands
  Machine Learning Engineer Intern – Advanced Concepts Team — Sep. 2025 – Nov. 2025
  — Optimized solar sail trajectories with PPO for low-thrust, long-horizon missions, emphasizing constraint handling and simulation-driven training. Improved transfer times by 20% compared to state-of-the-art genetic baselines.
- **BMW Group** — Munich, Germany
  Machine Learning Engineer Intern – Research & Development, Aerodynamics — Jan. 2025 – Jul. 2025
  — Researched GNNs and Graph Transformers for next-generation vehicle drag prediction, with emphasis on tire-shape modeling. Optimized for sub-second inference with 1M+ mesh points on a single H100 GPU.
  — Built pipelines for 3D scan acquisition, experiment management, and distributed training/inference.

Research Projects

- **Inductive Moment Matching for Discrete Language Diffusion** — Lausanne, Switzerland
  CLAIRE Laboratory – Prof. Caglar Gulcehre — Jun. 2025 – Nov. 2025
  — Adapted Inductive Moment Matching to discrete language diffusion by replacing Gaussian/Laplacian kernels in the MMD loss with information-diffusion kernels. Optimized training and evaluation for dLLMs with 100M+ parameters.
- **Diffusion Based Counterfactual Attacks** — Lausanne, Switzerland
  Computer Vision Laboratory – Prof. Pascal Fua, Mathieu Salzmann — Sep. 2024 – Jan. 2025
  — Developed adversarial attacks using DDPM-generated counterfactuals to stress-test classifier robustness, implementing adjoint-sensitivity backpropagation through the full stochastic diffusion process.
- **Reinforcement Learning for Design Optimization** — Lausanne, Switzerland
  Laboratory for Applied Mechanical Design – Prof. Jürg Alexander Schiffmann — Sep. 2024 – Jan. 2025
  — Optimized multi-component actuator designs with PPO, finding Pareto-efficient solutions with fewer samples than genetic optimizers.

Honors

— **Excellence Fellowship**: scholarship awarded by EPFL to the top 3% of incoming Master's students.
— **Best Freshman Prize**: scholarship awarded by Politecnico di Milano for the highest GPA.
— **Gold Medal** at the International Chemistry Olympiad, IChO 2019.
— **Gold Medal** at the National Chemistry Olympiad, 2019.
— **Bronze Medal** at the Italian Physics Olympiad, 2018 and 2019.

Skills, Activities & Interests

Languages: English (fluent), Italian (native). GRE: (VR: 162, Q: 170, AW: 5.0).
Extracurricular: EPFL Changemakers (entrepreneurship program), EPFL Rocket Team (R&D), Skyward Experimental Rocketry (R&D), Fundamentals of Quantum Computing.
Technical Skills: Python, C++, Linux, PyTorch, Ansys Fluent, Abaqus, SolidWorks.
Interests: Math Puzzles, Competitive Programming, Aviation, 3D Printing, Italian Cooking, Running.